

Properties of Series Representations

In the first example, $e^{1/z}$, we used an already known series and composition of functions to discover the relevant expansion without performing integration. This will be very useful, but relies on some properties of power series.

Thm: If a power series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ converges at a point $z=z_1$, then it is absolutely convergent for all z s.t. $|z-z_0| \leq |z_1-z_0|$. Moreover, the series is uniformly convergent for all z s.t. $|z-z_0| < |z_1-z_0|$.

A series is absolutely convergent if its series of moduli

$$\sum_{n=0}^{\infty} |a_n z^n| = \sum_{n=0}^{\infty} |a_n| |z|^n$$

is convergent. This property ensures that the series can be summed in any order, but also is the basis of many other properties.

Uniform convergence is a property that requires convergence be independent of z in some sense. We will not focus on the specifics here. More detail can be found in Brown-Churchill.

We are interested in ways of finding series representations, ideally without explicitly computing the integral formulas. To do this we need to know that Taylor series are unique when they converge, and hence that we can combine and manipulate them.

Thm: Let $\{z: |z-z_0| < R\}$ be the circle of convergence of the series

$$S(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n.$$

Also let C be a closed contour inside the circle and $g(z)$ be continuous on C . Then:

■ $S(z)$ is a continuous function of z

■ $\int_C g(z) S(z) dz = \sum_{n=0}^{\infty} a_n \int_C g(z) (z-z_0)^n dz$ (integration term-by-term)

■ $S(z)$ is analytic at each point z in the circle of convergence

■ The derivative of $S(z)$ has a series representation

$$S'(z) = \sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1} \quad (\text{differentiation term-by-term})$$

■ The series is the Taylor series ie. $a_n = \frac{1}{n!} S^{(n)}(z_0)$ for all n .

(uniqueness of series representations)

Similarly these results hold for the full Laurent series inside their annulus of convergence.

Example: $f(z) = \frac{e^z - 1}{z}$

At first this function appears to be singular at $z=0$.

However, in fact

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + \sum_{n=1}^{\infty} \frac{z^n}{n!} \quad \text{for all } z \in \mathbb{C}$$

$$\Rightarrow e^z - 1 = \sum_{n=1}^{\infty} \frac{z^n}{n!} \quad \text{for all } z \in \mathbb{C}$$

$$\Rightarrow \frac{e^z - 1}{z} = \sum_{n=1}^{\infty} \frac{z^{n-1}}{n!} = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} z^n = 1 + \frac{1}{2}z + \frac{1}{6}z^2 + \dots$$

The series converges for all $z \in \mathbb{C}$ and so we can choose $f(0) = 1$ to make f an entire function.

It is also possible to multiply and divide series so long as they converge both converge.

Thm: Suppose $f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$ and $g(z) = \sum_{n=0}^{\infty} b_n (z-z_0)^n$ converge within the circle $|z-z_0|=R$. Then for all z in $|z-z_0| < R$ the series

$$f(z)g(z) = \sum_{n=0}^{\infty} c_n (z-z_0)^n \quad \text{where} \quad c_n = \sum_{k=0}^n a_k b_{n-k}$$

also converges.

$$= \sum_{k=0}^n \binom{n}{k} f^{(k)}(z_0) g^{(n-k)}(z_0)$$

Example: $f(z) = \frac{e^z}{1+z} = \left[\sum_{n=0}^{\infty} \frac{z^n}{n!} \right] \cdot \left[\sum_{n=0}^{\infty} (-1)^n z^n \right]$

$$= \left(1 + z + \frac{1}{2}z^2 + \frac{1}{6}z^3 + \dots \right) \left(1 - z + z^2 - z^3 + \dots \right)$$

$$= \begin{aligned} &1 - z + z^2 - z^3 + \dots \\ &+ z - z^2 + z^3 - \dots \\ &+ \frac{1}{2}z^2 - \frac{1}{2}z^3 + \dots \\ &+ \frac{1}{6}z^3 + \dots \end{aligned}$$

$$= 1 + \frac{1}{2}z^2 - \frac{1}{3}z^3 + \dots$$

A similar formula can be found for division, although it is more complicated. If $f(z)/g(z)$ is analytic on the disc $|z-z_0| < R$ then its Taylor series converges and

$$\frac{f(z)}{g(z)} = \sum_{n=0}^{\infty} d_n (z-z_0)^n$$

and d_n can be found by differentiating f/g .

Example: $\frac{1}{z^2 \sinh z} = \frac{1}{z^2 \left(z + \frac{z^3}{3!} + \frac{z^5}{5!} + \dots \right)}$

$$= \frac{1}{z^3} \frac{1}{1 + \frac{1}{3!}z^2 + \frac{1}{5!}z^4 + \dots}$$

$$\begin{aligned}
&= \frac{1}{z^3} \left(1 - \frac{\left(\frac{1}{3!} z^2 + \frac{1}{5!} z^4 + \dots \right)}{1 + \frac{1}{3!} z^2 + \frac{1}{5!} z^4 + \dots} \right) \\
&= \frac{1}{z^3} \left(1 - \frac{1}{3!} z^2 \left(\frac{1 + \frac{3!}{5!} z^2 + \dots}{1 + \frac{1}{3!} z^2 + \frac{1}{5!} z^4 + \dots} \right) \right) \\
&= \frac{1}{z^3} \left(1 - \frac{1}{3!} z^2 \left(1 + \left(\frac{3!}{5!} - \frac{1}{3!} \right) z^2 + \dots \right) \right) \\
&= \frac{1}{z^3} \left(1 - \frac{1}{3!} z^2 + \left(\frac{1}{5!} - \frac{1}{(3!)^2} \right) z^4 + \dots \right) \\
&= \frac{1}{z^3} \left(1 - \frac{1}{6} z^2 + \frac{7}{360} z^4 + \dots \right) \\
&= \frac{1}{z^3} - \frac{1}{6z} + \frac{7}{360} z + \dots
\end{aligned}$$

Isolated Singular Points

The Cauchy-Goursat Theorem says that

- if f is analytic inside a simple closed contour C then

$$\int_C f(z) dz = 0$$

- if f is analytic inside a simple closed contour C except at a finite number of isolated points z_k then

$$\int_C f(z) dz = \sum_{k=0}^N \int_{C_k} f(z) dz$$

where C_k is a contour around the single singularity z_k .

Moreover, we now know, in a small neighbourhood of an isolated singularity z_k , the function f has a Laurent expansion

$$f(z) = \sum_{n=1}^{\infty} \frac{b_n}{(z-z_k)^n} + \sum_{n=0}^{\infty} a_n (z-z_k)^n$$

where $b_n = \frac{1}{2\pi i} \int_{C_k} f(z) (z-z_k)^{n-1} dz$.

In particular, $b_1 = \frac{1}{2\pi i} \int_{C_0} f(z) dz$.

Hence, the Laurent series can be used to determine the value of integrals of f .

Def: A singularity of f , z_0 , is isolated if f is analytic in the punctured disc $\{z: 0 < |z - z_0| < \delta\}$ for some $\delta > 0$.

The coefficient $b_1 = \frac{1}{2\pi i} \int_{C_0} f(z) dz$ of the Laurent series

where C_0 is a simple closed contour around z_0 , is the Residue of f at z_0 , $\text{Res}_{z=z_0} f(z)$.

This leads us to Cauchy's Residue Theorem.

Thm: (Cauchy's Residue)

Let C be a simple closed contour oriented positively and suppose f is analytic on an inside C except at a finite number of singularities z_k . Then

$$\int_C f(z) dz = 2\pi i \sum_{k=1}^N \text{Res}_{z=z_k} f(z)$$

Example: Compute $\int_C \frac{5z-2}{z(z-1)} dz$ where C is the circle of radius 2 oriented positively.

We need to compute the Laurent series expansions around $z=0, 1$.

$$\begin{aligned} \blacksquare \quad \frac{5z-2}{z(z-1)} &= \frac{5z-2}{z} \cdot \frac{-1}{1-z} = -\left(5 - \frac{2}{z}\right) (1 + z + z^2 + \dots) \\ &= \frac{2}{z} - 3 - 3z + \dots \end{aligned}$$

$$\begin{aligned} \blacksquare \quad \frac{5z-2}{z(z-1)} &= \frac{5z-2}{z-1} \cdot \frac{1}{1+(z-1)} = \left(5 + \frac{3}{z-1}\right) (1 - (z-1) + (z-1)^2 - \dots) \\ &= \frac{3}{z-1} + 2 - 2(z-1) \end{aligned}$$

Hence,
$$\int_C \frac{5z-2}{z(z-1)} dz = 2\pi i (2+3) = 10\pi i.$$

Example: Note that $z=0$ is not an isolated singularity of $\text{Log } z$ due to the presence of the branch cut.

Moreover, if $z(t) = Re^{it}$ for $t \in [-\pi, \pi]$ then

$$\begin{aligned} \int_C \text{Log}(z) dz &= \int_{-\pi}^{\pi} \text{Log}(Re^{it}) iRe^{it} dt \\ &= \int_{-\pi}^{\pi} (\ln R + it) iRe^{it} dt \\ &= \underbrace{iR \ln R \int_{-\pi}^{\pi} e^{it} dt}_{=0} - \underbrace{R \int_{-\pi}^{\pi} te^{it} dt}_{=-2\pi i R} \\ &= -2\pi Ri \end{aligned}$$

depends on R and so we do not expect to find a Laurent series for $\text{Log } z$ at $z=0$.

Example: Compute $\int_C z^2 \sin\left(\frac{1}{z}\right) dz$ where C is the positively oriented unit circle.

The integrand is analytic for all $z \neq 0$. So we find the Laurent series at $z=0$.

$$\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}$$

$$\Rightarrow \sin\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{z^{2n+1}}$$

$$\Rightarrow z^2 \sin\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{z^{2n-1}} = z - \frac{1}{3!z} + \frac{1}{5!z^3} - \dots$$

The residue is $\text{Res}_{z=0} f(z) = -\frac{1}{3!} = -\frac{1}{6} \Rightarrow \int_C z^2 \sin\left(\frac{1}{z}\right) dz = -\frac{\pi i}{3}.$

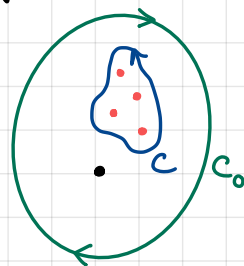
Residue at Infinity

It is sometimes convenient to think of the point at infinity as a singularity of an analytic function.

Note we have shown that if f is analytic and not constant, it cannot be bounded. Hence, f will generically have a singularity at infinity.

We can find the residue associated with this singularity by integration.

Def: Suppose f is analytic on the entire complex plane except for a finite number of singularities which are contained inside a simple closed contour C oriented positively. Let C_0 be a circle of radius R_0 oriented negatively and chosen to contain C .



$$\text{Then } \text{Res}_{z=\infty} f(z) = \frac{1}{2\pi i} \int_{C_0} f(z) dz \quad \left(= -\frac{1}{2\pi i} \int_C f(z) dz \text{ by deformation} \right).$$

We can also find this residue by the Laurent series.

Suppose we know the Laurent series at $z=0$ for f . Then by analyticity for $|z| > R_0$ we have

$$f(z) = \sum_{n=-\infty}^{\infty} c_n z^n \quad \text{for } R_1 < |z| < \infty.$$

$$\text{where } c_n = \frac{1}{2\pi i} \int_{C_0} \frac{f(z)}{z^{n+1}} dz \quad \text{and} \quad c_{-1} = -\text{Res}_{z=0} f(z)$$

$$\text{But then } f\left(\frac{1}{z}\right) = \sum_{n=-\infty}^{\infty} c_n \frac{1}{z^n} = \dots + c_2 \frac{1}{z^2} + c_1 \frac{1}{z} + c_0 \cdot 1 + \underbrace{c_{-1} z + \dots}_{\text{Residue on wrong power}}$$

$$\begin{aligned} \Rightarrow \frac{1}{z^2} f\left(\frac{1}{z}\right) &= \sum_{n=-\infty}^{\infty} c_n \frac{1}{z^{n-2}} \\ &= \dots + c_2 \frac{1}{z^4} + c_1 \frac{1}{z^3} + c_0 \frac{1}{z^2} + c_{-1} \frac{1}{z} + \dots \end{aligned}$$

$$\text{ie. } -\text{Res}_{z=0} f(z) = c_{-1} = \text{Res}_{z=0} \left[\frac{1}{z^2} f\left(\frac{1}{z}\right) \right]$$

Corollary: If f is analytic everywhere except a finite number of singular points interior to a positively oriented simple closed contour C , then:

$$\int_C f(z) dz = 2\pi i \operatorname{Res}_{z=0} \left[\frac{1}{z^2} f\left(\frac{1}{z}\right) \right].$$